This is one of my two preparation documents for my oral exam, focused on geometric/combinatorial group theory.

Preliminaries

Group-Theoretic Constructions

There are two major ways we may construct new groups out of old: extensions/semidirect products and pushouts/amalgamated free products. We will provide a brief overview of both of these.

Definition: Let Q and N be groups. We say a group G is an extension of Q and N if there is an exact sequence

$$1 \to N \to G \to Q \to 1.$$

If the extension splits — i.e., there is a section $Q \to N$ — then $G \cong Q \ltimes_{\varphi} N$ is a semidirect product for some $\varphi\colon Q \to \operatorname{Aut}(N)$.

Example:

- If the homomorphism $\varphi\colon Q \to \operatorname{Aut}(N)$ is trivial, then $N \rtimes_{\varphi} Q \cong N \times Q$ is a direct product.
- The map $\mathbb{Z}/2\mathbb{Z} \to \operatorname{Aut}(\mathbb{Z}/n\mathbb{Z})$ given by inversion yields the dihedral group

$$D_n \cong \mathbb{Z}/2\mathbb{Z} \ltimes \mathbb{Z}/n\mathbb{Z}.$$

- If G is any group, the lamplighter group of G is the semidirect product $\mathbb{Z} \ltimes G^{\mathbb{Z}}$ with homomorphism $\varphi\colon \mathbb{Z} \to \operatorname{Aut}\left(G^{\mathbb{Z}}\right)$ given by

$$\varphi(z) = ((g_n)_{n\in\mathbb{Z}} \mapsto (g_{n+z})_{n\in\mathbb{Z}}).$$

- the wreath product of G and H, denoted $G \wr H$, is the semidirect product $H \ltimes G^{H}$ with $\varphi\colon H \to \operatorname{Aut}\left(G^{H}\right)$ given by the shift action

$$\varphi(h) = \left((g_k)_{k\in H} \mapsto (g_{kh})_{k\in H}\right).$$

Definition: Let A be a group, $\alpha_1 : A \to G_1$, and $\alpha_2 : A \to G_2$ group homomorphisms. We say a group G is a *pushout* of G_1 and G_2 over A with respect to α_1 and α_2 if there are homomorphisms $\beta_1 : G_1 \to G$ and $\beta_2 : G_2 \to G$ satisfying $\beta_1 \circ \alpha_1 = \beta_2 \circ \alpha_2$ and the universal property for pushouts.

That is, for every group H and all group homomorphisms $\varphi_1 : G_1 \to H$ and $\varphi_2 : G_2 \to H$ satisfying $\varphi_1 \circ \alpha_1 = \varphi_2 \circ \alpha_2$, there is exactly one homomorphism $\varphi\colon G \to H$ satisfying $\varphi \circ \beta_i = \varphi_i$ for each $i = 1, 2$. We denote the pushout by $G_1 *_A G_2$.

- If $A = \{e\}$ is the trivial group, then the pushout $G_1 *_A G_2 = G_1 * G_2$ is called the *free product* of G_1 and G_2.
- If α_1 and α_2 are injective, then $G_1 *_A G_2$ is called an amalgamated free product of G_1 and G_2 over A.

Example:

- The free group F_n on n letters is the n-fold free product $\mathbb{Z} * \cdots * \mathbb{Z}$.
- The infinite dihedral group $D_\infty \cong \mathbb{Z}/2\mathbb{Z} \ltimes \mathbb{Z}$ is isomorphic to the free product $\mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/2\mathbb{Z}$.
- The matrix group $\mathrm{SL}_2(\mathbb{Z})$ is isomorphic to the amalgamated free product $\mathbb{Z}/6\mathbb{Z} *_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$.
- If X and Y are pointed, non-empty, path-connected topological spaces with nonempty and path-connected intersection, then the fundamental group of their union (which is a type of pushout) is given by the amalgamated free product of their fundamental groups over the fundamental group of the intersection.

Theorem: Pushout groups exist and are unique up to isomorphism.

Proof. Take presentations

$$G_i = \left\langle \left\{ x_g : g \in G_i \right\} \middle| R_i \right\rangle.$$

Then, the presentation

$$G := \left\langle \left\{ x_g : g \in G_1 \right\} \cup \left\{ x_g : g \in G_2 \right\} \middle| \left\{ x_{\alpha_1(a)} x_{\alpha_2(a)}^{-1} : a \in A \right\} \cup R_1 \cup R_2 \right\rangle.$$

The group homomorphisms $\beta_i \colon G_i \to G$ taking $g \mapsto x_g$, along with the relations $\left\{ x_{\alpha_1(a)} x_{\alpha_2(a)}^{-1} : a \in A \right\}$, give $\beta_1 \circ \alpha_1 = \beta_2 \circ \alpha_2$. The universal property then follows from the universal property of generators and relations. □

Definition: Let G be a group, and let $A, B \subseteq G$ be isomorphic subgroups with isomorphism $\theta \colon A \to B$. The *HNN extension* of G with respect to θ is the group

$$G*_\theta := \left\langle \left\{ x_g : g \in G \right\} \cup \{t\} \middle| \left\{ t^{-1} x_a t x_{\theta(a)}^{-1} : a \in A \right\} \cup R \right\rangle,$$

where R is the relations for G. We say that t is the stable letter of the HNN extension.

The HNN extension is the universal (in a sense) way to force isomorphic subgroups to be conjugate to each other.

Group Homology/Cohomology

Definition: Let R be a commutative unital ring. The *group ring* is the ring $R[G]$ with

- underlying abelian group $\bigoplus_{g \in G} Rg$, where the g are formal symbols corresponding to the elements of G;
- multiplication given by the R-bilinear extension of the group multiplication $(g, h) \mapsto gh$.

If G is a non-abelian group and R is non-trivial, then the group ring $R[G]$ is not commutative.

Definition: Let G be a group.

- The *simplicial resolution* of G, denoted $C_*(G)$, is the chain complex of $\mathbb{Z}$-modules whose chains are defined by

  $$C_n := \bigoplus_{G^{n+1}} \mathbb{Z},$$

  with boundary maps given by $\partial_n \colon C_n(G) \to C_{n-1}(G)$ taking

  $$\partial_n \left(\sum_{g \in G^{n+1}} a_g (g_0, \dots, g_n) \right) = \sum_{g \in G^n} a_g \sum_{j=0}^{n} (-1)^j \left(g_0, \dots, g_{j-1}, g_{j+1}, \dots, g_n \right).$$

- If A is a left $\mathbb{Z}[G]$-module, then the simplicial chain complex of G with A-coefficients is given by $C_n(G; A) = A \otimes_{\mathbb{Z}[G]} C_*(G)$ with boundary operator $\partial_*^A = \mathrm{id}_A \otimes_{\mathbb{Z}[G]} \partial_*$. The elements of $\ker\left(\partial_n^A\right)$ are called the n-cycles.

  The n-th homology of G with A-coefficients is the $\mathbb{Z}$-module given by

  $$H_n(G; A) := \frac{\ker\left(\partial_n^A \colon C_n(G; A) \to C_{n-1}(G; A)\right)}{\operatorname{im}\left(\partial_{n+1}^A \colon C_{n+1}(G; A) \to C_n(G; A)\right)}.$$

- Let A be a left $\mathbb{Z}[G]$-module. The simplicial cochain complex of G with A-coefficients is defined by

  $$C^*(G; A) := \mathrm{Hom}_{\mathbb{Z}[G]}(C_*(G), A)$$

with coboundary operator ∂_A^* given by the A-dual of ∂_*. Elements of $\ker\left(\delta_A^n\right)$ are called the n-cocycles. The nth cohomology of G with A-coefficients is the $\mathbb{Z}$-module

$$H^n(G;A) := \frac{\ker\left(\partial_A^n \colon C^n(G;A) \to C^{n+1}(G;A)\right)}{\operatorname{im}\left(\partial_A^{n-1} \colon C^{n-1}(G;A) \to C^n(G;A)\right)}.$$

Nilpotent Groups

Lemma: Let G be a group, let $N \trianglelefteq G$ be a normal subgroup, and let $H, K, L \leq G$ be subgroups. If $[[H,K],L]$ and $[[K,L],H]$ are contained in N, then so is $[[L,H],K]$.

Proof. The subgroups $[[H,K],L]$, $[[K,L],H]$, and $[[L,H],K]$ are generated by the commutators $[h,k,\ell]$, $[k,\ell,h]$, and $[\ell,h,k]$ respectively, where $h \in H$, $k \in K$, and $\ell \in L$. The Hall–Witt identity says that

$$\begin{aligned} e &= b^{-1}\left[a, b^{-1}, c\right]b \\ &= c^{-1}\left[b, c^{-1}, a\right]c \\ &= a^{-1}\left[c, a^{-1}b\right]a, \end{aligned}$$

so all of these commutators are contained in N. $\square$

Definition: Let G be a group. The *lower central series* of G is the sequence $(\gamma_k(G))_{k\geq 1}$ of subgroups defined recursively by

$$\begin{aligned} \gamma_1(G) &:= G \\ \gamma_{k+1}(G) &:= [\gamma_k(G), G]. \end{aligned}$$

We say a group G is *nilpotent* if its lower central series is finite. That is, if there is some integer $k \geq 1$ such that $\gamma_k(G) = e$. The smallest such k is called the *nilpotence class* (or nilpotency class) of G.

Inductively, we observe that $\gamma_k(G)$ is characteristic in G with $\gamma_{k+1}(G) \subseteq \gamma_k(G)$, meaning that the lower central series is a descending normal series.

Lemma: Let G be a group. Then,

$$\left[\gamma_i(G), \gamma_j(G)\right] \subseteq \gamma_{i+j}(G).$$

Proof. We prove by induction on i. By the definition of γ_{i+1}, we have that $\left[\gamma_1(G), \gamma_j(G)\right] = \left[G, \gamma_j(G)\right] = \gamma_{j+1}(G)$.

Now, we show that

$$\left[\gamma_{i+1}(G), \gamma_j(G)\right] \subseteq \gamma_{i+j+1}(G).$$

First, we claim that given $a, b \in \gamma_{i+1}(G)$ and $c \in \gamma_j(G)$, we have

$$\begin{aligned} [ab, c] &= [a,c][b,c] \bmod \gamma_{i+j+1}(G) \\ \left[a^{-1}, c\right] &= [a,c]^{-1} \bmod \gamma_{i+j+1}(G). \end{aligned}$$

Since $a \in \gamma_{i+1}(G) \subseteq \gamma_i(G)$, the induction hypothesis has $[a,c] \in \left[\gamma_i(G), \gamma_j(G)\right] \subseteq \left[\gamma_{i+j}(G)\right]$. Consequently, $[[a,c],b] \in \left[\gamma_{i+j}(G), \gamma_j(G)\right] \subseteq \left[\gamma_{i+j}(G), G\right] = \gamma_{i+j+1}(G)$.

By standard commutator identities, we have

$$\begin{aligned} [ab, c] &= [a,c][a,c,b][b,c] \\ &= [a,c][b,c]\left([b,c]^{-1}[a,c,b][b,c]\right). \end{aligned}$$

Since $\gamma_{i+j+1}(G)$ is normal, it follows that $[b,c]^{-1}[a,c,b][b,c] \in \gamma_{i+j+1}(G)$, so this gives our first desired result. Similarly, by taking $b = a^{-1}$, we obtain the second desired result.

Therefore, we only need to show that $[a,x,b] \in \gamma_{i+j+1}(G)$ For all $a \in \gamma_i(G)$, $x \in G$, and $b \in \gamma_j(G)$.

From the Hall–Witt identity, we may write

$$[a,x,b] = x^{-1}\left(\left(a^{-1}\left[b,a^{-1}x^{-1}\right]a\right)^{-1}\left(b^{-1}\left[x^{-1},b^{-1}a\right]b\right)^{-1}\right)x.$$

while by the inductive hypothesis, $\left[b,a^{-1},x^{-1}\right] \in \left[\gamma_{i+j}(G),G\right] = \gamma_{i+j+1}(G)$, and $\left[x^{-1},b^{-1},a\right] \in \left[\gamma_{j+1}(G),\gamma_i(G)\right] \subseteq \gamma_{i+j+1}(G)$, seeing as $\gamma_{i+j+1}(G)$ is a normal subgroup (hence the entire conjugacy class must land in it if any conjugate lands in it). □

We say a descending series $G = G_1 \geq G_2 \geq \cdots \geq G_k \geq G_{k+1} \geq \cdots$ is *central* if $\left[G_i,G_j\right] \subseteq G_{i+j}$ for all $i,j \geq 1$. This condition implies that each G_i is normal in G.

Lemma: Let G be a group. Suppose $(G_k)_{k\geq 1}$ is a descending central series for G. Then, $G_i \supseteq \gamma_i(G)$ for each $i \geq 1$.

Proof. We proceed by induction. First, $G_1 = G = \gamma_1(G)$.

Suppose the statement holds for i. The defining property of a central series and the inductive hypothesis give

$$\begin{aligned} G_{i+1} &\supseteq [G_i,G] \\ &\supseteq [\gamma_i(G),G] \\ &= \gamma_{i+1}(G). \end{aligned}$$

□

In particular, this means that a group G is nilpotent if and only if it admits any finite descending central series.

Proposition: Let G be a nilpotent group with nilpotence class c. Then, any subgroup and any quotient of G is nilpotent of nilpotence class at most c.

Proof. If $H \leq G$ is a subgroup, inductively we have $\gamma_i(H) \subseteq \gamma_i(G)$.

If $\varphi\colon G \to \overline{G}$ is a surjective homomorphism, then we have $\varphi([H,K]) = [\varphi(H),\varphi(K)]$ for all subgroups $H,K \leq G$. In particular, this means that inductively we have $\gamma_{i+1}(\varphi(G)) = \varphi(\gamma_{i+1}(G))$. □

Definition: Let G be a group. The *upper central series* of G is the ascending series $(Z_k(G))_{k\geq 0}$, where

$$\begin{aligned} Z_0(G) &= e \\ Z_{i+1}(G) &= \pi_i^{-1}(Z(G/Z_i(G))), \end{aligned}$$

where $\pi_i\colon G \to G/Z_i(G)$ is the projection homomorphism, and $Z(H)$ is the center of a group H.

In particular, we have $Z(G/Z_i(G)) = Z_{i+1}(G)/Z_i(G)$.

If G admits a finite upper central series, then the descending central series $(Z_{k-i+1})_{i=1}^{k+1}$ is central.

Lemma: Let G be a group. Suppose G admits a finite central series $G = G_1 \geq G_2 \geq \cdots \geq G_r = \{e_G\}$. Then, $Z_i(G) \supseteq G_{r-i}$ for all $i \geq 0$. In particular, this means $Z_{r-1}(G) = G$.

Proof. We induct on i. First, observe that $Z_0(G) = \{e_G\} = G_r$.

We will show that $Z_{i+1}(G) \supseteq G_{r-i-1}$. That is, we have to show that $\pi_i(G_{r-i-1}) \subseteq Z(G/Z_i(G))$. This is equivalent to showing that $[G_{r-i-1},G] \subseteq Z_i(G)$. Yet, this follows from the fact that the sequence $(G_i)_{i\geq 0}$ is central, whence $[G_{i-r-1},G] \subseteq G_{r-i}$, and $G_{r-i} \subseteq Z_i(G)$ by the induction hypothesis. □

In particular, observe that $\gamma_i(G) \subseteq Z_{c-i}(G)$ if G is nilpotent of nilpotence class c.

Finitely Generated Nilpotent Groups

We start by considering torsion-free finitely generated nilpotent groups. That is, finitely generated nilpotent groups where no nontrivial element has finite order.

Lemma: Let G be a torsion free (not necessarily finitely generated) nilpotent group with upper central series $(Z_k(G))_{k\geq 0}$. Then, all the factors $Z_{i+1}(G)/Z_i(G)$ are torsion-free abelian groups.

Proof. We use induction on nilpotence class. If $c = 1$, then $Z_1(G)/Z_0(G) = G$ is abelian, so it is a torsion-free abelian group by assumption.

Suppose it holds true for all nilpotent groups of nilpotence class $(c-1)$. Let G have nilpotence class c. Set $H = G/Z_1(G)$. Then, we have $Z_i(H) = Z_{i+1}(G)/Z_i(G)$ by induction and the definition of the upper central series. In particular, H has nilpotence class $(c-1)$.

By the third isomorphism theorem, we have $Z_{i+1}(G)/Z_i(G) \cong (Z_{i+1}(G)/Z_1(G))/(Z_i(G)/Z_1(G)) \cong Z_i(H)/Z_{i-1}(H)$, so it suffices to show that H is torsion-free to complete the proof.

Let $a \in G$. Suppose there exists an integer $m \geq 1$ such that $a^m \in Z_1(G)$, and suppose toward contradiction that $a \notin Z_1(G)$. Notice that $Z_1(G) = Z(G)$, so by assumption there is some $b \in G$ such that $[a,b] \neq e$. Let $1 \leq i \leq c-1$ be the maximal index for which there exists some $b \in \gamma_i(G)$ such that $[a,b] \neq e$, where $(\gamma_i(G))_{i=1}^{c+1}$ is the lower central series. By the various commutator identities, we have

$$\begin{aligned} e &= [a^m, b] \\ &= \left[aa^{m-1}, b\right] \\ &= a^{1-m}[a,b]a^{m-1}\left[a^{m-1}, b\right]. \end{aligned}$$

Since $[a,b] \in \gamma_{i+1}(G)$, we must have that a commutes with $[a,b]$, so $a^{1-m}[a,b]a^{m-1} = [a,b]$, and we have $e = [a^m, b] = [a,b]\left[a^{m-1} b\right]$. Iterating this process gives $e = [a,b]^m$, but since G is torsion free, this gives that $[a,b] = e$, which is a contradiction of the definition of b. Thus, $a \in Z_1(G)$, hence $H = G/Z_1(G)$ is torsion-free. ☐

We will use this to prove Malcev's Theorem. Given a unital (commutative) ring R, we define the upper unitriangular matrices $\mathrm{UT}(n,R)$ to be the set of all upper triangular matrices with 1 along the diagonal.

Theorem (Malcev's Theorem)**:** Let G be a finitely generated torsion-free nilpotent group. Then, there exists $n \geq 1$ and an embedding $G \to \mathrm{UT}(n,\mathbb{Z})$.

We start by reducing to the case of showing for $\mathrm{UT}(n,\mathbb{Q})$. This follows from the fact that if one is given an embedding $\iota\colon G \to \mathrm{UT}(n,\mathbb{Q})$, and S is the finite generating set for G, we may take N to be the least common multiple of the denominators of the entries of the elements of $\iota(S)$. Conjugating each element

$$\iota(s) = \begin{pmatrix} 1 & & a_{i,j} \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

by the matrix

$$M_N = \mathrm{diag}\left(1, N, \dots, N^{n-1}\right),$$

we obtain an upper unitriangular matrix with entries $\beta_{i,j} = N^{-i}\alpha_{i,j}N^j$, which is contained in $\mathbb{Z}$ since $i < j$. If $\gamma_{M_N} \in \mathrm{Aut}(\mathrm{UT}(n,\mathbb{Q}))$ is conjugation by this matrix, it follows that $\gamma_{M_N} \circ \iota$ gives the desired embedding into $\mathrm{UT}(n,\mathbb{Z})$.

We next recall some results from representation theory.

Definition: If G is a group and $\mathbb{K}$ is a field, then a $\mathbb{K}$-vector space V equipped with a linear action G — i.e., an algebra homomorphism $\mathbb{K}[G] \to \mathrm{End}_{\mathbb{K}}(V)$ — is called a G-module.

The G-equivariant linear maps (also known as G-endomorphisms) $\varphi\colon V \to V$ form a $\mathbb{K}$-subalgebra of $\mathrm{End}_{\mathbb{K}}(V)$, and are denoted by $\mathrm{End}_G(V)$. The G-invariant subspaces of a G-module V are called G-submodules of V, and we say V is irreducible if the only proper G-submodule is the trivial submodule.

Lemma (Schur's Lemma)**:** Let V be an irreducible G-module, and let $\varphi \in \operatorname{End}_G(V)$ be a G-endomorphism of V. If φ admits an eigenvector, then φ is a scalar multiple of the identity map $\mathrm{id}_V \colon V \to V$.

Definition: A linear map $\varphi \colon V \to V$ is said to be *unipotent* if 1 is the only eigenvalue.

If V is finite-dimensional, then Cayley–Hamilton immediately gives that φ is unipotent if and only if $(\varphi - \mathrm{id}_V)^n = 0$.

Lemma: Let V be a finite-dimensional vector space over a field $\mathbb{K}$. Let $G \subseteq \operatorname{End}_{\mathbb{K}}(V)$ be a unipotent nilpotent subgroup. Then, there exists a basis for V such that $G \subseteq \mathrm{UT}(n, \mathbb{K})$.

Notice here that we want to find a single basis such that *all* the elements of G act as upper unitriangular matrices (i.e., this is not a tautological statement).

Proof. Assume G acts irreducibly. Suppose toward contradiction G is then nontrivial. By nilpotency, its center $Z(G) = Z_1(G)$ is also nontrivial. Let $z \in Z(G) \setminus \{\mathrm{id}_V\}$. Notice then that elements G then become $Z(G)$-module homomorphisms, since we have that $gz(v) = zg(v)$, so by Schur's lemma, z is a multiple of id_V. Since z is also unipotent, it must be the case that $z = \mathrm{id}_V$, which is a contradiction. Thus, $G = \mathrm{id}_V$, and since the action is irreducible, V is one-dimensional. ☐

Groups and Graphs

One of the first connections between groups and geometry is the notion of a Cayley graph, and the way that the action of a group on a graph is connected to the structure of the group.

Graph Theory Review

Definition:

- A graph is a pair $X = (V, E)$, where $E \subseteq V^{[2]} := \{e : e \subseteq V, |e| = 2\}$. Elements of E are denoted by $\{v_1, v_2\}$ if the graph is undirected. If the graph is directed, then elements of E are denoted by tuples (v_1, v_2).
- If (V, E) is a graph, then we say vertices $v, v' \in V$ are adjacent if $\{v, v'\} \in E$. The number of neighbors is the degree of the vertex.
- If $X = (V, E)$ and $X' = (V', E')$ are graphs, then we say X and X' are isomorphic if there is a graph isomorphism between X and X' — that is, a bijection $f \colon V \to V'$ such that for any $v, w \in V$, we have $\{v, w\} \in E$ if and only if $\{f(v), f(w)\} \in E'$.

Definition: Let $X = (V, E)$ be a graph.

- A path in X is a sequence $[v_0, \dots, v_n]$ of distinct vertices such that $\left\{v_j, v_{j+1}\right\} \in E$.
- A graph is called connected if every pair of its vertices can be connected by a path in X.
- A cycle of length n is a path $[v_0, \dots, v_n]$ such that $\{v_n, v_0\} \in E$.
- A tree is a connected graph without cycles. Note that this is equivalent to a graph where every pair of vertices has exactly one path connecting them.
- If X is a connected graph, a *spanning tree* for X is a tree that contains all the vertices of X.

In algebraic topology, we use spanning trees as a way to compute fundamental groups. This is done by using the fact that spanning trees are CW subcomplexes (viewing the graph as a CW complex).

Cayley Graphs

Definition: Let G be a group, and let $S \subseteq G$ be a generating set for G without the identity. The Cayley graph of G with respect to S is the graph $\mathrm{Cay}(G, S)$, with

- vertices given by elements of G;

- edges given by

$$\left\{\{g, gs\} : g \in G, s \in S \cup S^{-1}\right\}.$$

Example:

- The Cayley graph of $\mathbb{Z}$ with respect to $\{1\}$ appears as an infinite line, while the Cayley graph of $\mathbb{Z}$ with respect to the generating set $\{2, 3\}$ appears as a particular 4-regular graph. See [Löh17, Figure 3.3].
- The Cayley graph of $\mathbb{Z}^2 \subseteq \mathbb{R}^2$ with respect to the generating set $\{(1,0),(0,1)\}$ appears as the integer lattice.
- The Cayley graph of $\mathbb{Z}/6\mathbb{Z}$ with respect to $\{[1]\}$ is a cycle graph.
- The Cayley graph of the symmetric group S_3 with respect to the generating set $\{\sigma, \tau\}$, where we have

$$S_3 := \left\langle \sigma, \tau | \sigma^3, \tau^2, \tau\sigma\tau\sigma \right\rangle.$$

- The Cayley graph of any group with respect to the generating set given by the group itself is a complete graph.

One of the seemingly simple questions one might ask is when exactly finitely generated groups G and H with finite generating sets $S \subseteq G$ and $T \subseteq H$ are such that $\operatorname{Cay}(G, S) \cong \operatorname{Cay}(H, T)$ are isomorphic as graphs. This is essentially a question about Cayley graphs whose automorphisms are affine (i.e., given by a group isomorphism and translation by a fixed group element).

- Cayley graphs of finitely generated abelian groups satisfy the rigidity question; the automorphisms are affine on the free part, and Cayley graphs remember the rank of the group and the size of the torsion subgroup.
- Automorphisms of Cayley graphs of finitely generated torsion-free nilpotent groups are affine.
- Finitely generated free groups admit isomorphic Cayley graphs if and only if they have the same rank.

Given a presentation of a group, the presentation complex is a two-dimensional CW complex given by taking one zero-cell, adding circles for every generator, and attaching disks for every relation (attached by the word for the relation).

The presentation complex associated to $\left\langle x, y | xyx^{-1}y^{-1} \right\rangle$. The presentation complex associated to $\left\langle x | x^2 \right\rangle$ is $\mathbb{RP}^2$.

In general, any group admits a classifying space (known as the Eilenberg MacLane space) whose fundamental group is the underlying group and whose higher homotopy groups are trivial. These spaces are unique up to homotopy equivalence. For example, the torus is a classifying space for $\mathbb{Z}^2$ and $\mathbb{RP}^\infty$ is a classifying space for $\mathbb{Z}/2\mathbb{Z}$.

The 1-skeleton of the universal cover of the presentation complex of the presentation $\langle X | R \rangle$ is basically the Cayley graph (save for generators of order 2).

Theorem: Let F be a free group freely generated by $S \subseteq F$. Then, $\operatorname{Cay}(F, S)$ is a tree.

Proof. Every element of F is a unique word in elements of S, so there is a unique path between the identity and any word, so there is a unique path between any two elements, so $\operatorname{Cay}(F, S)$ is a tree. □

There is a converse, but it requires a caveat. In particular, we see that $\operatorname{Cay}(\mathbb{Z}/2\mathbb{Z}, [1])$ is a one-edge two-vertex graph (which is a tree).

Theorem: Let G be a group, $S \subseteq G$ a generating set satisfying $st \neq e$ for all $s, t \in S$. If $\operatorname{Cay}(G, S)$ is a tree, then $G \cong F(S)$ is freely generated by S.

Proof. Let S and G be such that $\operatorname{Cay}(G, S)$ is a tree. We will show that G is isomorphic to $F(S)$ by an isomorphism that is the identity on S.

First, we observe that, by the universal property applied to the inclusion $S \subseteq G$, there is a group homomorphism $\varphi\colon F(S) \to G$ that is the identity on S. Since S generates G, it follows that φ is surjective.

Assume toward contradiction that φ is not injective. Let $s_1 \cdots s_n \in F(S) \setminus \{\varepsilon\}$ be a word of minimal length mapped to $e \in G$ by φ.

First, since $\varphi|_S = \mathrm{id}_S$, we must have $n > 1$.

Next, if $n = 2$, then we would have

$$\begin{aligned} e &= \varphi(s_1 s_2) \\ &= \varphi(s_1)\varphi(s_2) \\ &= s_1 s_2, \end{aligned}$$

contradicting the fact that $s_1 \cdots s_n$ is reduced with $st \neq e$ in G.

Next, suppose $n \geq 3$. Inductively, we may consider the sequence of elements $g_0, \dots, g_{n-1}$ in G given by $g_{j+1} = g_j s_{j+1}$ for all $0 \leq j \leq n-2$. By minimality of the word $s_1 \cdots s_n$, the elements $g_0, \dots, g_{n-1}$ are distinct, and since $g_n = s_1 \cdots s_n = e$, it follows that $[g_0, \dots, g_{n-1}]$ is a cycle in $\mathrm{Cay}(G,S)$. Yet, this contradicts the fact that $\mathrm{Cay}(G,S)$ is a tree. Thus, φ is injective. □

It turns out that, generally speaking, we are able to reconstruct a Cayley graph from any action of a group on a graph.

Proposition: Let G be a group, and let G act on a connected graph $X = (V,E)$ by graph automorphisms. If this action is free and transitive on the set V of vertices of X, then for a distinguished vertex x, the set

$$S := \{s \in G : \{x, s \cdot x\} \in E\}$$

generates G, and $\mathrm{Cay}(G,S) \cong X$ are isomorphic as graphs.

Proof. Since the action of G on the vertices is free and transitive, we may identify V with G by the identification $g \mapsto g \cdot x$.

We start by showing that S generates G. Let $g \in G$. Since the graph X is connected, there is a path $[x_0, x_1, \dots, x_n]$, where $x_0 = x$ and $x_n = g \cdot x$, that joins x and $g \cdot x$. We will translate this path into steps in the group G as follows.

For each $0 \leq j \leq (n-1)$, define

$$g_j = \varphi^{-1}\left(x_j\right)$$

in G. Then, we take $s_j = g_j^{-1} g_{j+1}$. We will show that $s_j \in S$. Since $[x_0, \dots, x_n]$ is a path in X, we have $\left\{x_j, x_{j+1}\right\}$ is an edge of X. Since G acts by graph automorphisms, we have that $\left\{g_j^{-1} \cdot x_j, g_j^{-1} \cdot x_{j+1}\right\}$ is also an edge of X, but this is equal to $\left\{x, \left(g_j^{-1} g_{j+1}\right) \cdot x\right\}$. In particular, this means that

$$\begin{aligned} g &= g_n \\ &= g_0 g_0^{-1} g_1 g_1^{-1} \cdots g_{n-1} g_{n-1}^{-1} g_n \\ &= s_0 s_1 \cdots s_{n-1}, \end{aligned}$$

so S is a generating set for G.

We only need to show now that $\mathrm{Cay}(G,S)$ is isomorphic to the graph X. We will show that φ induces a graph isomorphism. If $g, h \in G$, then we observe that $\{\varphi(g), \varphi(h)\} = \{g \cdot x, h \cdot x\}$. Since G acts by graph automorphisms on X, the set is an edge of X if and only if $\left\{x, \left(g^{-1} h\right) \cdot x\right\}$ is an edge of X, which holds if and only if $g^{-1} h \in S$, which is equivalent to $\{g, h\}$ being an edge of $\mathrm{Cay}(G,S)$. Thus, $\mathrm{Cay}(G,S)$ is isomorphic to X. □

Actions of Groups on Trees

Similar to how a group is a free group with generating set S if and only if $\text{Cay}(G,S)$ is a tree, there is a converse direction. That is, a group is free if and only if it admits a free action on a non-empty tree.

One direction emerges immediately from the fact that left-translation is a free action of G on itself that acts on its Cayley graph.

The reverse direction is a little more difficult. If $G \curvearrowright T$ is an action, we will have to find a tree that is a Cayley graph for G.

Definition: Let G be a group that acts on a connected graph X by graph automorphisms. A *spanning tree* of this action is a subgraph of X that is both a tree and contains exactly one vertex of every orbit of the action of G on the vertices of X.

Theorem: Every action of a group on a connected graph by automorphisms has a spanning tree.

Proof. Let G be a group acting on a connected graph X. We may assume that X is nonempty.

Consider the set T_G of all subtrees of X that contain at most one vertex of every orbit. This is a nonempty set since X has at least one vertex. The set T_G is partially ordered by the subgraph relation, and every totally ordered chain of T_G has an upper bound in T_G (namely, the union of the trees in this chain). Thus, by Zorn's Lemma, there is a maximal element T in T_G that is necessarily nonempty since X is nonempty.

Suppose toward contradiction that T is not a spanning tree for the action of G. Then, there is a vertex v such that none of the vertices of the orbit $G \cdot v$ is a vertex of T. We will show that there is some such v that is adjacent to a vertex of T.

Since X is connected, there is a path p connecting some vertex u of T with v, which we will label as

$$p = [u, \dots, v', \dots, v],$$

where v' is the first vertex of p that is not in T.

(i) If none of the vertices of the orbit $G \cdot v'$ are contained in T, then v' is the vertex with our desired property.

(ii) If there is some $g \in G$ such that $g \cdot v'$ is a vertex of T, write a subpath $p' = [v', \dots, v]$ of p, and consider the path

$$g \cdot p' = [g \cdot v', \dots, g \cdot v].$$

Notice that $g \cdot v$ is not contained in T since none of the elements of the orbit of v are contained in T by assumption. Since the length of p' is smaller than the length of p, we may iterate this process to eventually obtain our desired vertex.

Having obtained our vertex v with none of the vertices of the orbit $G \cdot v$ in T and some neighbor u of v is contained in T, we may add the edge $\{u, v\}$ to T to produce a tree in T_G that contains T as a proper subgraph. This yields the contradiction, as we had assumed T was maximal. Thus, T is a spanning tree for the action of G on X. □

In order to show the reverse direction, we will have to determine, given an action $G \curvearrowright T$, a set S such that $\text{Cay}(G,S)$ is a tree.

Theorem: If a group G acts freely on a tree T, then G is a free group.

Proof. Let T' be a spanning tree for the action of T. Say two translations $g \cdot T'$ and $h \cdot T'$ are adjacent if $g \cdot T' \neq h \cdot T'$ and there exist vertices $v \in g \cdot T'$ and $w \in h \cdot T'$ such that $\{v, w\}$ is an edge of T. Define the graph X by saying $V(X)$ consists of the set $\{g \cdot T' : g \in G\}$ and $E(X)$ consists of all the adjacent trees $\{g \cdot T', h \cdot T'\}$.

We will start by showing that if $g \neq h$, then $g \cdot T' \neq h \cdot T'$. We may reduce this question to showing that for any $g \neq e$, we have that $g \cdot T' \neq T'$. Yet, this follows directly from our assumptions; there is exactly one representative from each orbit in the spanning tree, and we cannot have $g \cdot v = v$ for any $v \in T'$ unless $g = e$ by the assumption of the freeness of the action G.

For any adjacent $g \cdot T'$ and $h \cdot T'$, we claim that there is a unique connecting edge. We observe that $g \cdot T'$ and $h \cdot T'$ are adjacent if and only if $h^{-1} g \cdot T'$ is adjacent to T' — that is, there exists $u \in T'$ and $v \in h^{-1} g \cdot T'$ such that $\{u, v\}$ is an edge. Yet, since u is a unique representative of the orbit of v, it follows that there is only one such edge. (In Löh's book, such an edge is dubbed an "essential edge" on page 89.)

We observe that, since each copy of T' is a tree (hence connected), and T itself is a tree, it follows that there can only be one unique path between any two vertices in T' as else that implies the existence of two non-unique paths in the original graph T.

Finally, since $V(X) \curvearrowleft G$ freely by taking $(h \cdot T') \cdot g = (hg) \cdot T'$, we must have that $\mathrm{Cay}(G, S) \cong X$. Since X is a tree, it follows that G is a free group. $\square$

The following is a more topological proof that relies heavily on covering space theory.

Proof via Covering Spaces. Let $G \curvearrowright T$ freely by graph automorphisms. We may view T as a one-dimensional CW complex; since T is a tree, this one-dimensional CW complex is simply connected, so the covering map $T \to T/G$ has it such that T is the universal cover for the CW complex T/G. In particular, we have that the deck transformations of T are exactly G, and $\pi_1(T/G) \cong G$.

Since any one-dimensional CW complex is homotopy-equivalent to a bouquet of circles, the Seifert–Van Kampen theorem gives that $\pi_1(T/G) \cong F(S)$ for some generating set S, so that G is free. $\square$

We now focus on some applications.

Theorem (Nielsen–Schreier Theorem): Let G be a free group, and let $H \leq G$ be a subgroup. Then, H is a free group.

Proof. If $G \curvearrowright T$ acts freely, then so too does H. $\square$

Quasi-Isometry of Metric Spaces and Groups

Definition: Consider two metric spaces (X, d) and (Y, ρ), and a map $f \colon X \to Y$.

- We say f is an isometric embedding if for all $x, x' \in X$, $\rho(f(x), f(x')) = d(x, x')$.
- We say f is an isometry (or isometric isomorphism) if f is an isometric embedding, and there exists an isometric embedding $g \colon Y \to X$ satisfying $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$.
- We say f is a bilipschitz embedding if there is some constant $C > 0$ such that for all $x, x' \in X$,
$$\frac{1}{C} d(x, x') \leq \rho(f(x), f(x')) \leq C d(x, x').$$
- We say f is a bilipschitz equivalence if it is a bilipschitz embedding and there exists a bilipschitz embedding $g \colon Y \to X$ satisfying $f \circ g = \mathrm{id}_Y$, $g \circ f = \mathrm{id}_X$. The spaces X and Y are then called bilipschitz equivalent.
- The map f is called a (C, K)-quasi-isometric embedding if for all $x, x' \in X$, we have
$$\frac{1}{C} d(x, x') - K \leq \rho(f(x), f(x')) \leq C d(x, x') + K.$$
- The map $f' \colon X \to Y$ is said to have finite distance from f if there is some $K > 0$ such that for all $x \in X$, we have $\rho(f(x), f'(x)) \leq K$.
- The map f is called a quasi-isometry if there is a quasi-inverse quasi-isometric embedding $g \colon Y \to X$. That is, g is a quasi-isometric embedding with $g \circ f$ having finite distance from id_X and $f \circ G$ having finite distance from id_Y. We say X and Y are quasi-isometric, and write $X \sim_{\mathrm{QI}} Y$.

Growth types of Groups

Definition: Let G be a finitely generated group with symmetric generating set S. The *word length* of $g \in G$ with respect to S, denoted $\ell_S(g)$, is the smallest $n \geq 0$ such that g may be written as a product of n elements of S. That is,

$$\ell_S(g) = \min\{n : g = s_1 s_2 \cdots s_n \text{ with } s_i \in S \text{ and } n \geq 0\}.$$

The word metric is then

$$d_S(g, h) = \ell_S\left(g^{-1} h\right).$$

Proposition: The word metric is a left-invariant metric. That is, it satisfies the following:

(i) $d_S(g, h) = 0$ if and only if $g = h$;

(ii) $d_S(g, h) = d_S(h, g)$;

(iii) $d_S(g, h) \leq d_S(g, k) + d_S(k, h)$;

(iv) $d_S(g, h) = d_S(kg, kh)$.

Proof.

(i) We have that $\ell(g) = 0$ if and only if $g = e$ by the fact that any nontrivial element must be a nontrivial product of elements of S.

(ii) Notice that $\ell(g) = \ell\left(g^{-1}\right)$ since S is symmetric, so if

$$g = s_1 \cdots s_n$$

realizes $\ell(g)$, then

$$g^{-1} = s_n^{-1} \cdots s_1^{-1}$$

must realize $\ell\left(g^{-1}\right)$, as otherwise we would have a smaller representation for g in terms of a product of elements of S.

(iii) Observe that, by the existence of relations and word reductions, we have $\ell(gh) \leq \ell(g) + \ell(h)$.

(iv) We have

$$\begin{aligned} d(kg, kh) &= \ell\left((kg)^{-1}(kh)\right) \\ &= \ell\left(g^{-1} k^{-1} kh\right) \\ &= \ell\left(g^{-1} h\right) \\ &= d(g, h). \end{aligned}$$

□

Amenability of Groups

Four Perspectives on Soficity

We will work to understand soficity (a type of generalized approximation property for groups) with a variety of perspectives.

The Law of the Group von Neumann Algebra

This is the approach that my advisor, Ben Hayes, takes towards tracial von Neumann algebras, specialized to the case of groups.

A tracial von Neumann algebra is a pair (M, τ), where M is a von Neumann algebra and $\tau\colon M \to \mathbb{C}$ is a linear functional that satisfies:

- positivity: $\tau(x^*x) \geq 0$ for all $x \in M$;
- normalization: $\tau(1) = 1$;
- traciality: for all $x, y \in M$, $\tau(xy) = \tau(yx)$.

If I is any index set, an I-tuple $a \in M^I$ yields a $*$-algebra homomorphism from the space of all noncommutative $*$-polynomials, denoted $\mathbb{C}^*\langle (T_i)_{i\in I}\rangle$, given by $P(a) \in M$ for all $P \in \mathbb{C}^*\langle (T_i)_{i\in I}\rangle$.

The *law* of an I-tuple $a \in M^I$ is then a linear functional $\ell_a\colon \mathbb{C}^*\langle (T_i)_{i\in I}\rangle \to \mathbb{C}$ given by

$$\ell_a(P) = \tau(P(a)).$$

Notice that in the case of a single-element set for I and a normal or self-adjoint element for A, the law is given by integration with respect to the spectral measure of a,

$$\ell_a(P) = \int P\, d\mu_a.$$

In a sense, the law is a generalization of the spectral measure.

Now, consider a group G with finite symmetric generating set S. Then, in the group von Neumann algebra $L(G)$, there is a distinguished tuple $(\lambda_s)_{s\in S} \in L(G)^S$ given by the left-translation operators λ_s for each $s \in S$. The corresponding law is then given by

$$\ell_S(P) = \langle P((\lambda_s)_{s\in S})\delta_e, \delta_e\rangle.$$

We start by viewing the symmetric group $\operatorname{Sym}(n) \subseteq U(\mathbb{M}_n(\mathbb{C}))$ as permutation matrices. Then, any S-tuple $\sigma \in \operatorname{Sym}(n)^S$ has law given by

$$\ell_\sigma(P) = \operatorname{tr}_n(P((\sigma_s)_{s\in S})).$$

Notice that, given a word w in $\mathbb{F}_S$, which we can view as a monomial in $\mathbb{C}^*\langle (T_i)_{i\in I}\rangle$, we observe that the non-normalized trace is the cardinality of the fixed-point set, so that

$$\operatorname{tr}_n(w((\sigma_s)_{s\in S})) = \frac{1}{n}\left|\{j \in [n] : w((\sigma_s)_{s\in S})(j) = j\}\right|.$$

We then say that G is sofic if and only if there is a sequence of natural numbers $(k_n)_n$ and a corresponding sequence of S-tuples $(\sigma_n)_n$ with each $\sigma_n \in \operatorname{Sym}(k_n)^S$, such that the laws ℓ_{σ_n} converge weak* (i.e., pointwise) to the law ℓ_S.

Metric Ultraproducts

The original notion of an ultraproduct came from consideration of normed spaces and a development of nonstandard analysis.

Let E_α be a family of normed spaces indexed by a set A. We may form the ℓ^∞-direct product of the set $\{E_\alpha : \alpha \in A\}$ by taking

$$\prod_{\alpha\in A} E_\alpha := \left\{(x_\alpha)_{\alpha\in A} : x_\alpha \in E_\alpha \text{ and } \sup_\alpha \|x_\alpha\| < \infty\right\},$$

and we may equip it with a norm

$$\|x\| = \sup_{\alpha\in A}\|x_\alpha\|_\alpha.$$

Write $\mathcal{E}$ for the ℓ^∞-direct product.

Given a (non-principal) ultrafilter $\mathcal{U}$ on A, there is a way to define a "null subspace" by taking

$$\mathcal{N}_{\mathcal{U}} = \left\{x \in \mathcal{E} : \lim_{\mathcal{U}}\|x_\alpha\| = 0\right\},$$

where the limit along the ultrafilter $\mathcal{U}$ of a sequence $(y_\alpha)_\alpha$ is defined to be y if for all $\varepsilon > 0$, the set $\left\{\alpha : \left\|y_\alpha - y\right\| < \varepsilon\right\}$ is contained in $\mathcal{U}$. The subspace $\mathcal{N}_{\mathcal{U}}$ is closed. The metric ultraproduct of the family $(E_\alpha)_{\alpha\in A}$ is then given by

$$\prod_{\mathcal{U}} E_\alpha \coloneqq \mathcal{E}/\mathcal{N}_{\mathcal{U}},$$

with norm given by

$$\left\|[x]_{\mathcal{U}}\right\| = \lim_{\mathcal{U}}\|x_\alpha\|.$$

We write E for the ultraproduct. When the ultrafilter is nonprincipal, then the ultraproduct is complete. If $(x_k)_k$ is a Cauchy sequence of elements of E, then for each $i \in \mathbb{N}$, we may find some N_i such that

$$\left\|x_N' - x_N\right\| < 2^{-i}$$

for all $N, N' \geq N_i$. For each k, we select a representative $(x_k(\alpha))_{\alpha\in A}$, and define

$$I_i = \left\{\alpha \in A : \left\|x_{N_i}(\alpha) - x_{N_{i+1}}(\alpha)\right\| < 2^{-i}\right\}.$$

We have that each I_i is contained in $\mathcal{U}$, and without loss of generality we may assume that $I_1 \supseteq I_2 \supseteq \cdots$ with empty intersection, since the ultrafilter $\mathcal{U}$ is non-principal, and thus is countably incomplete. Define an element $x \in \mathcal{E}$ by taking

$$x|_{I_i\setminus I_{i+1}} = x_{N_i}|_{I_i\setminus I_{i+1}}.$$

The equivalence class $[x]_{\mathcal{U}}$ is the limit of the Cauchy sequence $(x_k)_k$.

Next, we concern ourselves with ultraproducts of metric groups. Given a family (G_α, d_α) of topological groups with left-invariant metrics,[I] we may form an ultraproduct by following an analogous process for the case of metric spaces, but the result is only a metric space on which the Cartesian product group $\prod_{\alpha\in A} G_\alpha$ acts by isometries (i.e., a homogeneous space), rather than a proper group.

> **Example:** Consider the symmetric group $\operatorname{Sym}(\mathbb{N})$ on the natural numbers. The standard Polish topology on $\operatorname{Sym}(\mathbb{N})$ is the subspace topology induced by the product topology on the prodiscrete space $\mathbb{N}^{\mathbb{N}}$. This is a separable group topology with a compatible metric given by
>
> $$d(\sigma, \tau) = \sum_{i=1}^{\infty}\left\{2^{-i} : \sigma(i) \neq \tau(i)\right\}.$$
>
> We note that this is a bounded analogue of the Hamming distance for finite symmetric groups.
>
> We will form an ultrapower of the group $(\operatorname{Sym}(\mathbb{N}), d)$ with regard to a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. Since the metric d is itself bounded, we see that the space $\mathcal{G} = (\operatorname{Sym}(\mathbb{N}))^{\mathbb{N}}$ is the analogue of the space $\mathcal{E}$ from the construction of the ultraproduct of normed spaces.

[I] Such invariant metrics always exist for metrizable groups by a theorem of Kakutani.

> The set
>
> $$\mathcal{N} = \left\{x : \lim_{\mathcal{U}} d(x_\alpha, e) = 0\right\}$$
>
> is a subgroup of $\mathcal{G}$, seeing as the invariance of the metric d yields
>
> $$\begin{aligned} d(xy, e) &= d\left(y, x^{-1}\right) \\ &\leq d(y, e) + d\left(x^{-1}, e\right) \\ &= d(y, e) + d(x, e). \end{aligned}$$
>
> Yet, it is not normal. If we have transpositions $x = (x_i)_i = ((i, i+1))_{i\in\mathbb{N}}$ and $y = (y_i)_i = ((1, i))_{i\in\mathbb{N}}$, then $x \in \mathcal{N}$ but $y^{-1}xy \notin \mathcal{N}$, since $y_i^{-1}x_iy_i = (1, i+1)$ for each i, and we have $d\left(y_i^{-1}x_iy_i, e\right) \to 1/2$.
>
> In particular, this means that the homogeneous space $\mathcal{G}/\mathcal{N}$ admits a $\mathcal{G}$-invariant metric $d(x, y) = \lim_{\mathcal{U}} d(x_n, y_n)$, it is not a group.

In order to develop an ultraproduct of metric groups that is actually a group, we need to use bi-invariant metrics. That is, metrics that satisfy $d(x, y) = d(gx, gy) = d(xg, yg)$. These metrics are special because they admit topologies whose conjugation-invariant open subsets form a neighborhood basis at the identity.

If $(G_\alpha, d_\alpha)_\alpha$ are a family of metric groups with bi-invariant metrics, then the "null set" subgroup $\mathcal{N}$ is normal, and for any ultrafilter $\mathcal{U}$ on A, the ultraproduct

$$\prod_{\mathcal{U}} G_\alpha = \mathcal{G}/\mathcal{N}$$

is indeed a group. The bi-invariant metric

$$d(x\mathcal{N}, y\mathcal{N}) = \lim_{\mathcal{U}} d_\alpha(x_\alpha, y_\alpha)$$

yields the metric ultraproduct of the family (G_α, d_α) with respect to the ultrafilter $\mathcal{U}$.

> **Example** (Groups with Bi-Invariant Metrics):
>
> (i) If (X, μ) is a finite measure space, then group $\mathrm{Aut}(X, \mu)$ admits a bi-invariant metric given by
>
> $$d(\sigma, \tau) = \mu(\{x \in X : \sigma(x) \neq \tau(x)\}).$$
>
> (ii) The (normalized) Hamming distance on the symmetric group S_n is given by
>
> $$d_H(\sigma, \tau) = \frac{1}{n}|\{i \in [n] : \sigma(i) \neq \tau(i)\}|.$$
>
> This is the special case of a finite set $[n]$ equipped with the (normalized) counting measure.
>
> (iii) If H is a Hilbert space, then $U(H)$ denotes the group of unitary operators on H. The *uniform operator metric* is given by
>
> $$d_u(u, v) = \|u - v\|.$$
>
> This is a bi-invariant metric.
>
> (iv) If $H = \mathbb{C}^n$ is an n-dimensional Hilbert space, then $U(H) = U(n)$ is the unitary group of rank n. The *normalized Hilbert–Schmidt metric* on $U(n)$ is given by
>
> $$d_{\mathrm{HS}}(u, v) = \sqrt{\mathrm{tr}_n((u - v)^*(u - v))},$$
>
> where tr_n is the normalized trace $\mathrm{tr}_n = \frac{1}{n}\mathrm{Tr}$.

Every metric group G embeds into its ultrapower $G^{\mathcal{U}}$ with the diagonal embedding.

Approximate Homomorphisms

Cayley Graphs

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